

THERMAL ENGINEERING-I

Time Allowed: 3 Hours

Full Marks: 60

Answer the following questions from Group-A, B & C as directed.

Group A

1. Choose the correct answer from the alternatives (any ten):

1 x 10

- i. Heat pump works on the principle of - a) Kelvin plank Statement, ~~b) Clausius statement~~, c) Zeroth law of thermodynamics, d) energy equation of flow process.
- ii. In steady flow process - ~~a) the mass flow rate is constant~~, b) the heat transfer rate is constant, c) the work transfer rate is constant, d) all of these.
- iii. Superheated vapour behaves - a) Exactly as gas, b) As steam, c) As ordinary vapour, d) Approximately as a gas.
- iv. In an economiser - a) Feed water is heated by the heat extracted from flue gas, b) Primary air is heated by the heat extracted from the flue gas, c) Suspended solids are separated from the flue gas using electrostatic forces, d) Feed water is pumped to the de-aerator.
- v. Dryness fraction of a steam containing 20% of water vapour by mass is - a) 0.2, b) 80, c) 0.8, d) 20.
- vi. Heat and work are - (a) intensive properties, (b) extensive properties, (c) point functions, ~~(d) path functions~~.
- vii. Throttling of steam during dryness fraction determination by throttling calorimeter is essentially - a) isenthalpic process, b) isentropic process, c) isochoric process, d) isobaric process.
- viii. A series of operations, which take place in a certain order and restore the initial condition is known as- ~~a) reversible cycle~~, b) irreversible cycle, c) thermodynamic cycle, d) None of these.
- ix. The value of C_p/C_v for air is - ~~a) 1~~, b) 1.4, c) 1.8, d) 2.3.
- x. Mollier chart is also known as - a) T-S diagram b) p-T diagram c) p-V diagram d) h-S diagram
- xi. Which of the following statements is wrong? - a) Mechanical draught reduces the height of chimney., b) Natural draught reduces the fuel consumption., c) A balanced draught is a combination of induced and forced draught., d) All of the statements.
- xii. For a reversible adiabatic process, the change in entropy is - a) minimum, b) maximum, c) unity, ~~d) zero~~.
- xiii. According to Clausius statement of the second law of thermodynamics, the coefficient of performance of a refrigerator is never- (a) infinite, (b) greater than unity, (c) unity, ~~(d) less than unity~~.
- xiv. A device used in boiler to control the flow of steam from the boiler to the main pipe and shut off the steam completely when required, is known as - a) blow off cock, b) fusible plug, ~~c) stop valve~~, d) economiser.
- xv. Water tubes in a Babcock and Wilcox boiler are - ~~a) Horizontal~~, b) Vertical, c) Inclined, d) inclined.

2. Fill in the blanks (any ten):

1x10

- i. PMM2 violates _____ law.
- ii. Work is a _____ function.
- iii. Quantity of steam generated by a boiler at full load is called _____ of the boiler.
- iv. COP of heat pump is _____ than COP of refrigerator.
- v. First law of thermodynamics refers to the conservation of _____.
- vi. Thermal efficiency of a Carnot engine working between temperature levels T_1 and T_2 ($T_1 > T_2$) is given by _____.
- vii. Change of internal energy is zero during _____ process.
- viii. The unit of pressure in S.I. unit is _____.
- ix. One example of vertical fire tube boiler is _____.
- x. In La-mont boiler there is _____ circulation of water and steam.
- xi. For reversible adiabatic process _____ remains constant.
- xii. At critical point, the enthalpy of vaporisation i.e h_{fg} , is equal to _____.
- xiii. Total heat before throttling is _____ to total heat after throttling.
- xiv. As per Charles law, volume of given mass of a perfect gas varies _____ with its absolute temperature, when the absolute pressure remains constant.
- xv. Absolute zero temperature is _____.

3. Answer the following questions (any ten):

1x10

- i. State Boyle's law.
- ii. What is meant by quasi-static process in thermodynamics?
- iii. Differentiate universal gas constant and characteristic gas constant.
- iv. Give two examples of boiler accessories.
- v. Define Evaporation Ratio of a boiler.
- vi. Give one example of an open system.
- vii. Define Dryness fraction.
- viii. State Avogadro's law.
- ix. What is an isolated system?
- x. Write the value of characteristic gas constant?
- xi. What is superheated steam?
- xii. Under what circumstances, steam can be treated as an ideal gas?
- xiii. Why work is treated as high grade energy?
- xiv. Define a pure substance?
- xv. State the use of a calorimeter.

Group B

4. Answer the following questions (any six):

2x6

- i. Calculate the work done by a gas in an isothermal process from A to B. $P_A = 1 \text{ Pa}$, $V_A = 3 \text{ m}^3$ and $P_B = 3 \text{ Pa}$.
- ii. How does fusible plug work in fire tube boiler?
- iii. Explain in brief the concept of flow work.
- iv. What do you mean by entropy of wet steam?
- v. State Kelvin Planck statement.
- vi. Briefly state the function of an air-preheater.
- vii. Show different thermodynamic processes on P-V and T-S diagram.
- viii. Define a perfect gas.
- ix. Why circulating pump is required for high pressure boilers?
- x. What is latent heat of vaporisation?

Group C

5. Answer the following question (any one):

1x6

- i. A Carnot heat engine receives heat from a reservoir at 900°C at a rate of 800 kJ/min and reject the waste heat to the ambient air at 27°C . The entire work output of the engine is used to drive a refrigerator that removes heat from the refrigerated space at -5°C and transfers it to the same ambient air at 27°C . What will be the maximum rate of heat removal (in kJ/min) from the refrigerated space?
6
- ii. Define the following with examples: a) Open system, b) Closed system, c) Isolated system.
2+2+2
- iii. State First Law of Thermodynamics. What are the limitations of this law? Give an example of application of the law.
2+3+1

6. Answer the following question (any one):

1x6

- i. Show that $C_p - C_v = R$.
6
- ii. Prove that during adiabatic process, work done is given by $W = (P_1 V_1 / \gamma - 1) \{1 - P_2 V_2 / P_1 V_1\}$.
6
- iii. Steam expands isentropically in a nozzle from 1 MPa , 250°C to 10 kPa . The steam flow rate is 1 kg/s . Neglecting the K.E of steam at inlet to the nozzle, find the velocity of steam at exit from the nozzle and the exit area of the nozzle?
6

7. Answer the following question (any one):

1x6

- i. In a separating and throttling calorimeter the total quantity of steam passed was 40 kg and 2.2 kg of water was collected from separator. Steam pressure before throttling was 1.5 MPa and temperature and pressure after throttling are 120°C and 105 kPa . Determine the dryness fraction of steam before entering to calorimeter. Specific heat of superheated steam may be considered as 2.09 kJ/kg.K .
6
- ii. Draw a simple neat sketch of Cochran boiler, Label all its components and briefly mention some common features of the boiler.
6
- iii. What are boiler mountings and accessories? What is the difference between them? Give 3 examples each of boiler mountings and accessories.
2+2+2